

Ethan Allen-Fernandez

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EDUCATION

Rochester Institute of Technology (RIT)

*Bachelor of Science, **Software Engineering & Economics** Double Major*

Most recent semester GPA: 3.87

SWE Courses: Engineering Secure Software, Engineering Subsystems, Web Engineering, Algorithms.

Economics Courses: Econometrics, International Trade & Finance, Macroeconomics, Math for Economics.

Rochester, NY

Expected May 2026

SKILLS

- Programming Languages: Python (Proficient), C++ (Proficient), Java (Proficient), Typescript/Javascript (Proficient), HTML/CSS (Proficient), SQL (Proficient), C# (Prior Experience), C (Prior Experience), GO (Prior Experience), Rust (Prior Experience)
- Frameworks: Angular, ReactJS, NextJS
- Databases: PostgreSQL, MSSQL
- Experience in embedded, network, object-oriented, query, markup, and scripted programming
- Fluent in Spanish

EXPERIENCE

MOOG

Buffalo, NY/Remote

Software Engineer, Co-op/Part-Time

June 2025 – PRESENT

- Developed Python-based full-stack services supporting automated manufacturing workflows, improving system reliability and operator visibility.
- Designed and implemented a file transfer process for machinery used in automated manufacturing.
- Designed and developed the data collection and prediction process for determining when the manufacturing process will fail.
- Developed the process that checks the status of tool components used within machinery.

JCS Controls

Rochester, NY

Full Stack Developer, Co-op

January – August 2024, January – June 2025

- Built a full stack web application for customers to view and edit recipes and ingredients.
- Initially developed as an intern-led prototype; later promoted to a production system and commissioned as part of a \$20M+ industrial automation project.
- Part of a two person intern team during initial development, later given full development responsibility.
- Stack: MSSQL, .NET API, NextJS, Typescript, Tailwind CSS

PROJECTS

Startup - Dementia Assistance, Senior Project

August 2025 – PRESENT

- Developed a full stack application to help dementia patients reorient themselves when they wake up.
- Consists of a caregiver portal to upload and schedule custom media modules and individual patient applications that display the scheduled media.
- Focus on modular design to allow caregivers maximum flexibility based on individual care plans.
- Main responsibilities include database development, API development, patient device networking implementation, and module scheduling implementation.

HTTP2.5, Personal Project

November 2025 – PRESENT

- Designed and implemented a custom HTTP protocol variant over the SCTP transport protocol in C++.
- Implemented my own user-space SCTP stack via UDP Encapsulation (using reference [RFC9260]).
- Implemented HTTP in accordance with the semantics outlined in references [RFC9110] and [RFC9112].

Player Analytics Service (Soccer), Personal Project

December 2024 – March 2025

- Built a web scraper in Rust to gather player data.
- Built a request tunneling service in Python for the web scraper in order to bypass IP blocking and site scraping restrictions.
- Developed a k-mean cluster machine learning model in Python to categorize players into archetypes.

Embedded Music Player, Class Project

August – December 2023

- Coded a Nucleo-L476RG Board into a music player similar to an iPod using C.
- Played music via an attached speaker by parsing a collection of MIDI files.
- Played, paused, and skipped songs via an attached button.